SPECIFICATIONS FOR APPROVAL

DOCUMENT	(●) Preliminary Specification (　　) Final Specification
제 조 사	유비벨록스㈜
제 품 명	UONE-Q09
고객사 Part #	
제 정 일 자	2026년 03월 12일
부품 사진	[TOP　VIEW]　　　[BOTTOM　VIEW]

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[Document Revision History]

Date	담당부서	내용	비고

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Date	담당부서	내용

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1. Introduction

이 문서에서 기술하는 UONE-Q09은 AEC-Q100 grade2 규격을 만족하는 전장용 무선충전기用 eSE 규격에서 요구하는 기능을 포함하고, PKI를 사용한 기기 인증과 보안 저장소로서의 용도로 사용한다.

번호	항목	설명
1	Wafer fabrication site	N/A
2	Package site	N/A
3	Test/Final site	Jin-Chun, Republic of Korea, Ubivelox Jin-Chun Factory
4	ESD characteristics	3.1. 전기적 특성 참조
5	Silicon technology	65nm technology
6	Package feature	
7	MSL	LEVEL 1
8	General Information	. Chip model : SLC52AIA300N4 . CPU : 16bit . Pin Count : 10 . RAM: 12KB . Flash : 300KB . Supply Voltage: 1.62V to 5.5v . External Clock: up to 10 MHz . Internal Clock: up to 85 MHz . OS : Proprietary OS . Body Size : 3mm * 3mm * 0.55mm . Die Attach : . Wire Bonding : . Encapsulant : Mold

▶ **Process Site**

Process / Site	1st Case	2nd Case	3rd Case
Fab			
Assembly			
Final Test			

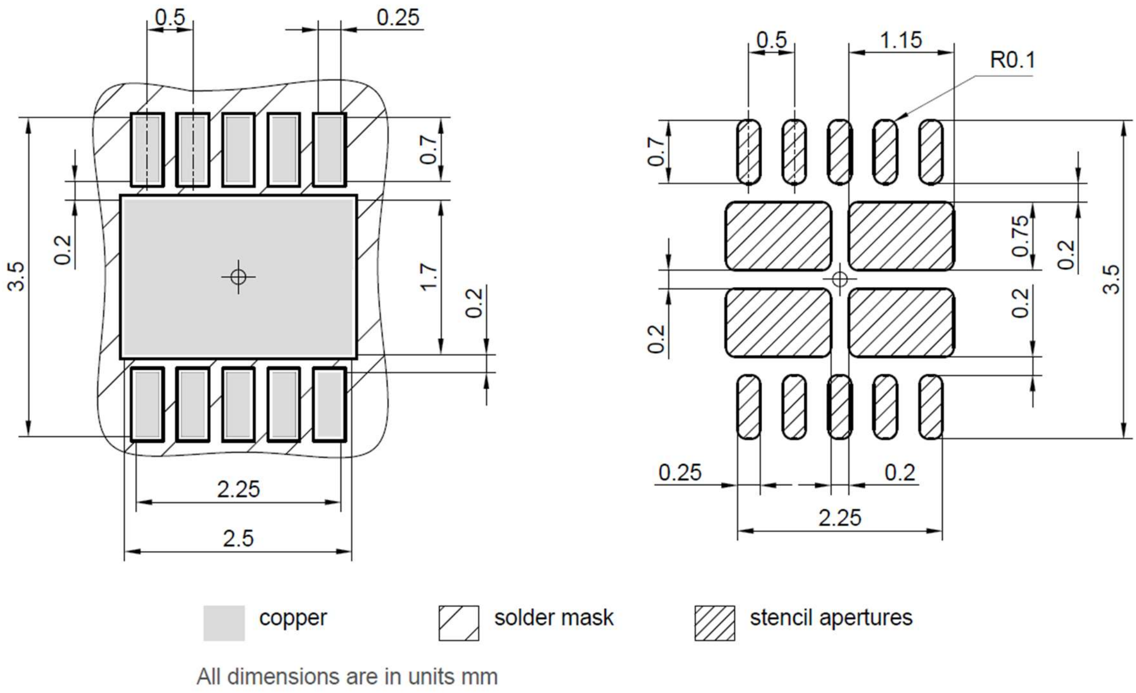
2. 형상

2.1 DFN10P 3x3 Package Information

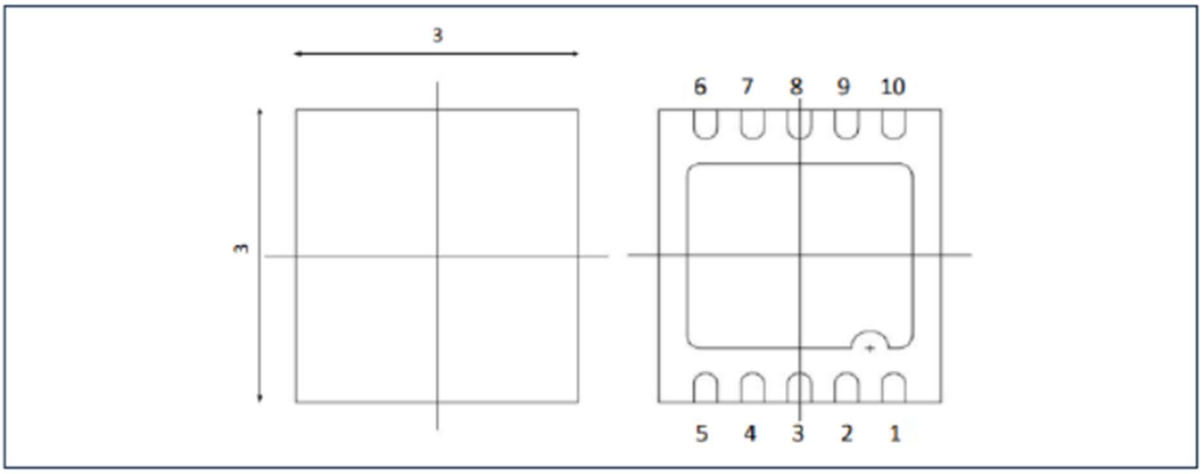
2.1.1 Outline Information

PKG SIZE			PKG Weight (Finished product)
X	Y	Height	
3.00±0.10	3.00±0.10	0.55±0.05	mg

- Foot print



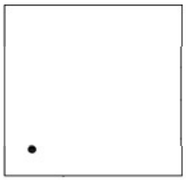
- Outline Drawing



- Pad to signal reference for PG-USON-10-package

Pad	Symbol	Pin type	Buffer type	Signal function / remarks
01	GND	GND	-	Power supply: Common ground reference (VSS)
02	NC			No internal connection / do not connect externally
03	GPIO0.1	I/O	GPIO_IO	**Software I/O (SWIO) usage:** GPIO0.1 **ISO/IEC 7816-3 card usage:** UART_IO **I2C usage (all modes):** SDA
04	NC			No internal connection / do not connect externally
05	NC			No internal connection / do not connect externally
06	NC			No internal connection / do not connect externally
07	NC			No internal connection / do not connect externally
08	GPIO0.0	I/O	GPIO_IO	**Software I/O (SWIO) usage:** GPIO0.0 **ISO/IEC 7816-3 card usage:** UART_CLK **I2C usage (clock-stretching mode):** SCL
09	GPIO0.2	I	GPIO_I	**Software I/O (SWIO) usage:** GPIO0.2
				ISO/IEC 7816-3 card usage: UART_RST **I2C usage (normal mode):** SCL
10	Vcc	PWR	-	Power supply: Power and pad supply (VCC)

- Top view Guide (Laser Marking) - **TBD**

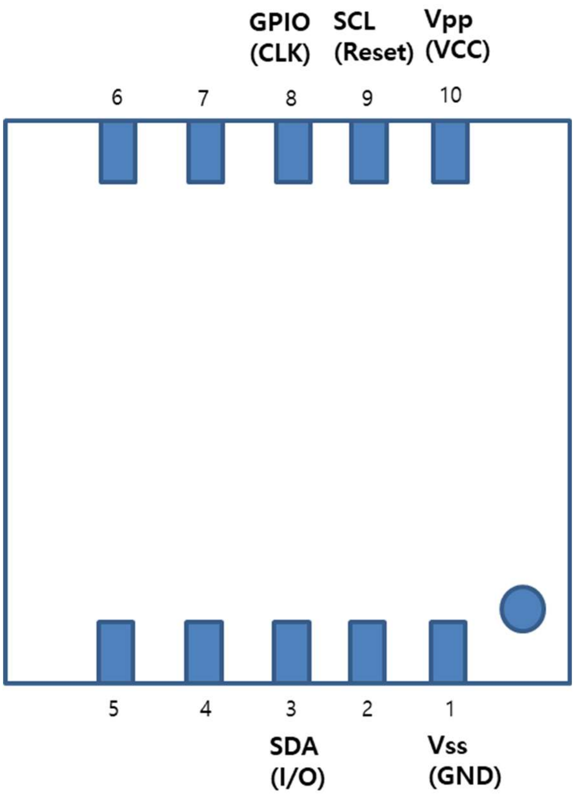


1st line : UONE-Q09(Product name) - TBD

2nd line : YWW(Year & Week code) - TBD

3rd line : • (Pin1)

- Pinout Information



No	Pin	No	Pin
1	Vss(GND)	6	NC
2	NC	7	NC
3	SDA	8	GPIO
4	NC	9	SCL
5	NC	10	Vpp

- Pin Descriptions

Signal	Description	
	I2C	**ISO7816**
Vss(GND)	Common ground reference	Ground supply (GND has to be connected to the main mother board ground)
SCL(Reset)	I2C serial clock Line Pin	External reset (Reset used to re-initialize the device)
SDA(I/O)	I2C serial data Line Pin	Input/output (ISO7816 IO0 data pin depending on the selected configuration)
Vpp(VCC)	Power supply pin	Power supply
GPIO(CLK)	General Purpose Input & Output PIN (Used for MCU interrupts.)	External Clock

3. DATASHEET

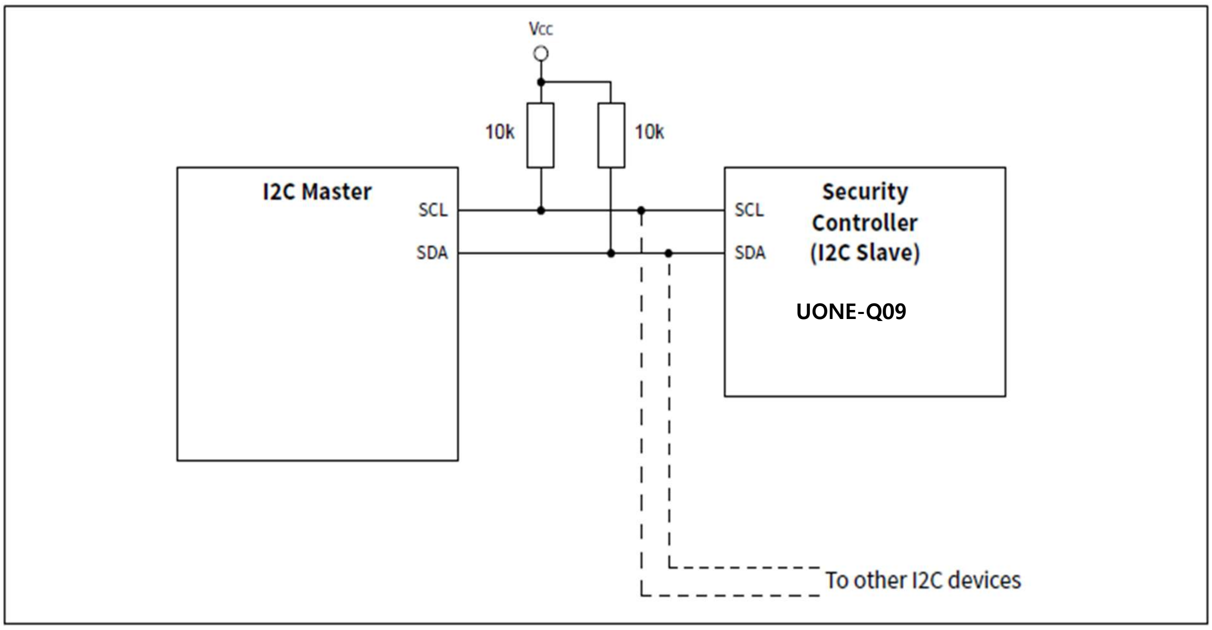
3.1 전기적 특성

3.1.1 Absolute Maximum ratings

Symbol	Parameter	Value	Unit
V_{CC}	Supply voltage	–0.3 to 7.0	V
V_{IN_GPIO}	GPIO with ISA feature	–0.3 to 7.0	V
T_A	Ambient operating temperature	–40 to +105	°C
T_j	Junction temperature	Max 115	°C
T_{STG}	Storage temperature (Please refer to package specification)	–65 to +150	°C
V_{ESD}	Electrostatic discharge voltage according to HBM	4000	V
	Electrostatic discharge voltage according to CDM	750	V

3.1.2 Recommended power supply filtering

- UONE-Q09 reference circuit for I^2C



3.1.3 DC characteristics

- DC Electrical Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Supply voltage	V_{CC}	1.62	–	5.5	V	Overall functional range
	$V_{CC\_ISO\_A}$	4.5	5.0	5.5	V	Supply voltage range for operation of ISO/IEC 7816-3 Class A
	$V_{CC\_ISO\_B}$	2.7	3.0	3.3	V	Supply voltage range for operation of ISO/IEC 7816-3 Class B
	$V_{CC\_ISO\_C}$	1.62	1.8	1.98	V	Supply voltage range for operation of ISO/IEC 7816-3 Class C
	V_{CC_GPIO}	1.62	–	5.5	V	Supply voltage range for operation of GPIO
	V_{CC_I2C}	1.62	–	5.5	V	Supply voltage range for operation of I2C
Supply current, no current limitation mode	I_{CCAVG}	–	10.5	–	mA	While running a Dhrystone test T_A = 25 °C; V_{CC} = 3.3V; f_{sys} = 85
Supply current, no current limitation mode	I_{CCAVG}	–	9.0	–	mA	While running a typical authentication profile[1)] T_A = 25 °C; V_{CC} = 5.0V; f_{sys} = 50MHz
Supply current, no current limitation mode	I_{CCAVG}	–	14.0	–	mA	While running a typical authentication profile[1)] T_A = 25 °C; V_{CC} = 5.0V; f_{sys} = 85 MHz
Supply current spikes[2)]	I_{CCD}	–	–	100	mA	Q ≤ 20 nAs; V_{CC} = $V_{CC\_ISO\_A}$
Supply current spikes[2)]	I_{CCD}	–	–	50	mA	Q ≤ 10 nAs; V_{CC} = $V_{CC\_ISO\_B}$
Supply current spikes[2)]	I_{CCD}	–	–	30	mA	Q ≤ 6 nAs; V_{CC} = $V_{CC\_ISO\_C}$
Supply current, in current limitation mode	I_{CCLIMA}	–	–	10	mA	V_{CC} = $V_{CC\_ISO\_A}$
Supply current, in current limitation mode	I_{CCLIMB}	–	–	6	mA	V_{CC} = $V_{CC\_ISO\_B}$
Supply current, in current limitation mode	I_{CCLIMC}	–	–	4	mA	V_{CC} = $V_{CC\_ISO\_C}$
Supply current, in sleep mode	I_{CCS1}	–	–	200	µA	T_A = 25 °C; f_{UART_CLK} = 1 MHz; all other GPIO inputs at V_{CC}, no other interface activity
Supply current, in sleep mode	I_{CCS2A}	–	–	200	µA	T_A = 25 °C; V_{CC} = $V_{CC\_ISO\_A}$; f_{UART_CLK} stopped; all other GPIO inputs at V_{CC}, no other interface activity

- Electrical Characteristics(Continued)

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Supply current, in sleep mode	I_{CCS2B}	–	–	100	µA	T_A = 25 °C; $V_{CC} = V_{CC\_ISO\_B}$; f_{UART_CLK} stopped; all other GPIO inputs at V_{CC}, no other interface activity
Supply current in sleep mode	I_{CCS2C}	–	–	100	µA	T_A = 25 °C; $V_{CC} = V_{CC\_ISO\_C}$; f_{UART_CLK} stopped; all other GPIO inputs at V_{CC}, no other interface activity
Supply current, in sleep mode	I_{CCS3}	–	–	100	µA	T_A = 25 °C; V_{CC_I2C} = 3.3 V; I2C ready for operation (no bus activity), all other GPIO inputs at V_{CC}, no other interface activity
Temperature sensor accuracy	T_{S_ACC}			± 6	°C	T_J = 25 °C to 110 °C

1) NVM read and write operations, SCP and Crypto active and operating in parallel to the CPU
2) The maximum spike amplitude and spike charge defined by ETSI TS 102 221 are kept within the specified range. The maximum spike length which is technically irrelevant for terminal regulator and voltage stability is typically kept within the specified range.

Note: In order to reduce the sleep power consumption to a minimum, the target setting for all sleep modes (unless otherwise specified) is PSU_SLEEP.LP_SLEEP set and PSU_SLEEP.CAT_SLEEP cleared.

3.1.4 AC characteristics

- AC Electrical Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
V_{CC} rampup time	t_{VCCR}	1	–	–	µs	0 to 100% of VCC target voltage ramp
BOS startup time until entry into user mode.	t_{WA}	–	–	–		Refer to corresponding Product List
NVM power-down time	t_{EEPD}	–	11	–	µs	T_A = 25 °C
NVM wake-up time[1]	t_{EEREC}	–	21	–	µs	T_A = 25 °C; NVM in CB mode (see NVMCONF2.CBCLMODE), f_{SYSCLK} = 20 MHz[2]
		–	38	–	µs	T_A = 25 °C; NVM in CL mode (see NVMCONF2.CBCLMODE), f_{SYSCLK} = 10 MHz[2]
Wake-up from sleep mode to initial CPU activity	t_{WUPS}	–	–	50	µs	
Wake-up from catnap sleep mode to initial CPU activity	t_{WUP_CN}	–	–	1	µs	
System oscillator frequency range	f_{OSC}	-	85	-	MHz	Average system frequency at 25 °C junction temperature, zero lifetime (without silicon aging). The actual oscillator frequency may vary due to dynamic on chip power management functions. The dynamic worst case minimum frequency may be significantly below the minimum average oscillator frequency $f_{OSC_AV_MIN}$The dynamic maximum frequency may be higher than $f_{OSC\_AV\_MAX}$
System oscillator frequency temperature drift	f_{OSC_T}		- 95		kHz/ °C	Average system oscillator frequency drift as function of junction temperature
1 MHz clock frequency range	$f_{CLK1MHZ}$	0.90	1	1.10	MHz	After NVM osc. calibration
System clock frequency	f_{SYSCLK}	f_{OSC} / 32		f_{OSC}	MHz	

1) The NVM needs a time t_{EEPD} for power down; if sleep mode is requested and a wake-up condition is pending, the total time the NVM is not available is $t_{EEPD} + t_{EEREC}$.

2) A higher f_{SYSCLK} does not significantly shorten the resulting time.

3.1.5 NVM Programming Characteristics

Attention: The NVM Programming Characteristics in the table below provide the pure hardware timing. For overall software timings refer to the HSL (Hardware Support Library) user documentation.

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
NVM write single bit up to a double word (4 Bytes)	t_{WR_BT}	–	56	–	µs	Relevant only for Single Bit Write Areas. Done through write block operation. T_A = 25 °C; f_{SYSCLK} = 20 MHz[1)]; NVM in CB mode (see NVMCONF2.CBCLMODE).
		–	76	–	µs	Relevant only for Single Bit Write Areas. Done through write block operation. T_A = 25 °C; f_{SYSCLK} = 10 MHz[1)]; NVM in CL mode (see NVMCONF2.CBCLMODE).
NVM write block (32 Bytes)	t_{WR_BL}	–	114	–	µs	T_A = 25 °C; f_{SYSCLK} = 20 MHz[1)]; NVM in CB mode (see NVMCONF2.CBCLMODE).
		–	177	–	µs	T_A = 25 °C; f_{SYSCLK} = 10 MHz[1)]; NVM in CL mode (see NVMCONF2.CBCLMODE).
NVM write overhead	t_{WR_OH}	–	54	–	µs	Once overall in continuous mode, once per block in one-shot mode. T_A = 25 °C; f_{SYSCLK} = 20 MHz[1)]; NVM in CB mode (see NVMCONF2.CBCLMODE).
		–	89	–	µs	Once overall in continuous mode, once per block in one-shot mode. T_A = 25 °C; f_{SYSCLK} = 10 MHz[1)]; NVM in CL mode (see NVMCONF2.CBCLMODE).
NVM verify or erase check block (32 Bytes)	t_{VER_BL}	–	1	–	µs	T_A = 25 °C; f_{SYSCLK} = 20 MHz[1)]

NVM verify or erase check overhead	t_{VER_OH}	–	26	–	µs	Verify @ 1 hardread level or erase check (@ hardread). Once overall in continuous mode, once per block in one-shot mode. T_A = 25 °C; f_{SYSCLK} = 20 MHz[1] ; NVM in CB mode (see NVMCONF2.CBCLMODE).
		–	52	–	µs	Verify @ 1 hardread level or erase check (@ hardread). Once overall in continuous mode, once per block in one-shot mode. T_A = 25 °C; f_{SYSCLK} = 10 MHz[1]; NVM in CL mode (see NVMCONF2.CBCLMODE).
NVM erase page or sector	t_{ER}	1.2	–	7.3	ms	T_A = 25 °C; f_{SYSCLK} = 20 MHz[1]; NVM in CB mode (see NVMCONF2.CBCLMODE). For erase timings in different erase modes.
		1.4	–	8.5	ms	T_A = 25 °C; f_{SYSCLK} = 10 MHz[1]; NVM in CL mode (see NVMCONF2.CBCLMODE). For erase timings in different erase modes.
NVM erase disturb handling	t_{ER_EDH}	–	22+8*x	–	µs	T_A = 25 °C; f_{SYSCLK} = 20 MHz[1]; NVM in CB mode (see NVMCONF2.CBCLMODE). x = number of block pairs (=2*256bit).
		–	43+10*x	–	µs	T_A = 25 °C; f_{SYSCLK} = 10 MHz[1]; NVM in CL mode (see NVMCONF2.CBCLMODE). x = number of block pairs (=2*256bit).

1) A higher f_{SYSCLK} does not significantly shorten the resulting time

3.1.6 I2C Interface Characteristics

- General I2C Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Supply voltage	V_{CC_I2C}	1.62	–	5.5	V	
SDA, SCL input voltage	V_{IN_I2C}	- 0.3	–	V_{CC_I2C} + 0.5 or 5.5[1)]	V	V_{CC_I2C} is in the operational supply range.
		- 0.3	–	5.5	V	V_{CC_I2C} is switched off. See section "Attention" below.

1) Whichever is lower

Attention: The pins used for operating the I2C interface are provided with an "indirect supply avoidance" feature (ISA) which allows to switch off the supply voltage of the security controller regardless of the input voltage VIN_I2C at these pins and without drawing a significant pad input current.

For operation of the I2C interface, the electrical characteristics are compliant with the I2C bus specification rev4 for "standard-mode" (fSCL up to 100 kHz) and "fast-mode" (fSCL up to 400 kHz), with certain deviations as stated in the table below.

- I2C Standard mode Interface Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
SCL clock frequency	f_{SCL}	0	–	100	kHz	
Input low-level	V_{IL}	- 0.3	–	0.3 * V_{CC_I2C}	V	
Low-level output voltage	V_{OL1}	0	–	0.4	V	Sink current 3 mA; $V_{CC\_I2C} \geq 2.7$ V Sink current 2 mA; $V_{CC\_I2C} < 2.7$ V
Low-level output current	I_{OL}	3 2	–	–	mA	V_{OL} = 0.4 V; $V_{CC\_I2C} \geq 2.7$ V V_{OL} = 0.4 V; $V_{CC\_I2C} <$
Output fall time from V_{IHmin} to V_{ILmax} (at device pin)	t_{OF}	–	–	250	ns	$C_b \leq 400$ pF; $V_{CC\_I2C} \geq 2.7$ V $C_b \leq 200$ pF; $V_{CC\_I2C} < 2.7$ V
Capacitive load for each bus line	C_b	–	–	400 200	pF	$V_{CC\_I2C} \geq 2.7$ V $V_{CC\_I2C} <$

- I2C Fast mode Interface Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
SCL clock frequency	f_{SCL}	0	–	400	kHz	
Input low-level	V_{IL}	- 0.3	–	0.3 * V_{CC_I2C}	V	
Low-level output voltage	V_{OL1}	0	–	0.4	V	Sink current 3 mA; $V_{CC\_I2C} \geq 2.7$ V Sink current 2 mA; $V_{CC\_I2C} < 2.7$ V
Low-level output current	$I_{OL\,(0.4)}$	3 2	–	–	mA	$V_{OL} = 0.4$ V; $V_{CC\_I2C} \geq 2.7$ V $V_{OL} = 0.4$ V; $V_{CC\_I2C} < 2.7$ V
Output fall time from V_{IHmin} to V_{ILmax} (at device pin)	t_{OF}	20 * V_{CC_I2C} / 5.5 V[1)]	–	250	ns	$C_b \leq 400$ pF; $V_{CC\_I2C} \geq 2.7$ V $C_b \leq 200$ pF; $V_{CC\_I2C} < 2.7$ V
Capacitive load for each bus line	C_b	15[1)]	–	400 200	pF	$V_{CC\_I2C} \geq 2.7$ V $V_{CC\_I2C} <$

1) A min. capacitive load C_{bMin} is necessary to reach $t_{OF\ Min}$.

- I2C Fast mode Plus Interface Characteristics

For operation of the I2C interface, the electrical characteristics are compliant with the I2C bus specification rev4 for "fast mode plus " (fSCL up to 1 MHz), with certain deviations as stated in the table below.

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
SCL clock frequency	f_{SCL}	0	–	1000	kHz	
Input low-level	V_{IL}	- 0.3	–	0.3 * V_{CC_I2C}	V	
Low-level output voltage	V_{OL1}	0	–	0.4	V	Sink current 3 mA; $V_{CC\_I2C} \geq 2.7$ V Sink current 2 mA; $V_{CC\_I2C} < 2.7$ V
Low-level output current	I_{OL}	3 2	–	–	mA	$V_{OL} = 0.4$ V; $V_{CC\_I2C} \geq 2.7$ V $V_{OL} = 0.4$ V; $V_{CC\_I2C} <$
Output fall time from V_{IHmin} to V_{ILmax} (at device pin)	t_{OF}	20 * V_{CC_I2C} / 5.5 V[1)]	–	120	ns	$C_b \leq 150$ pF
Capacitive load for each bus line	C_b	15[1)]	–	150	pF	

1) A min. capacitive load C_{bMin} is necessary to reach $t_{OF\ Min}$.

3.1.7 GPIO Interface Characteristics

- GPIO operation Supply and Input Voltages

Notes
1. All currents flowing into the pad are considered to be positive.
2. TA as given for the controller's operating temperature range unless otherwise stated.

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Supply voltage	V_{CC_GPIO}	1.62	–	5.5	V	
GPIO pad input voltage	V_{IN_GPIO}	- 0.3	–	V_{CC_GPIO} + 0.3	V	V_{CC_GPIO} is in the operational supply range.
		- 0.3	–	5.5	V	V_{CC_GPIO} is switched off. See section "Attention" below.

Attention: The pins used for operating the GPIO interface are provided with an "indirect supply avoidance" feature (ISA) which allows to switch off the supply voltage of the security controller regardless of the input voltage VIN_GPIO at these pins and without drawing a significant pad input current.

- GPIO DC Electrical Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Input current, pull-up (weak) enabled, IO pad	I_{PUW_IO}	-3	–	-20	μA	$0 V \le V_{IN\_GPIO} \le V_{CC\_GPIO} - 0.5 V$
Input current, pull-up (weak) enabled, input pad	I_{PUW_IN}	-3	–	-20	μA	$0 V \le V_{IN\_GPIO} \le V_{CC\_GPIO} - 0.5 V; V_{CC\_GPIO} \le 3.6 V$
		-3	–	-35	μA	$0 V \le V_{IN\_GPIO} \le V_{CC\_GPIO} - 0.5 V; V_{CC\_GPIO} > 3.6 V$
Input current, pull-down (weak) enabled	I_{PDW}	3	–	20	μA	$0.5 V \le V_{IN\_GPIO} \le V_{CC\_GPIO}$
Pull-up (strong) resistance	R_{PUS}	2.5	–	5.5	kΩ	$0 V \le V_{IN\_GPIO} \le V_{CC\_GPIO} - 0.5 V$
Input leakage current, IO pad	I_{LI_IO}	- 2	–	2	μA	Pull-up/down off, output stage off; $0 V \le V_{IN\_GPIO} \le V_{CC\_GPIO}$
Input leakage current, input pad	I_{LI_IN}	- 2	–	2	μA	Pull-up/down off, $0 V \le V_{IN\_GPIO\_I} \le V_{CC\_GPIO}; V_{CC\_GPIO} \le 3.6 V$

		- 15	–	2	µA	Pull-up/down off, 0 V ≤ $V_{IN\_GPIO\_I}$ ≤ V_{CC_GPIO};V_{CC_GPIO} > 3.6V
Input low voltage	V_{IL}	- 0.3	–	0.3 * V_{CC_GPIO}	V	
Input high voltage	V_{IH}	0.7 * V_{CC_GPIO}	–	V_{CC_GPIO} + 0.3	V	
Output low voltage	V_{OL}	–	–	0.3	V	I_{OL} = 1 mA
		–	–	0.4	V	I_{OL} = 4 mA, V_{CC_GPIO} ≥ 2.7V
Output high voltage	V_{OH}	V_{CC_GPIO} - 0.3	–	–	V	I_{OH} = - 1 mA
		V_{CC_GPIO} - 0.4	–	–	V	I_{OH} = - 4 mA, V_{CC_GPIO} ≥ 2.7V
Input capacitance[1)]	C_{IN}	–	–	10	pF	

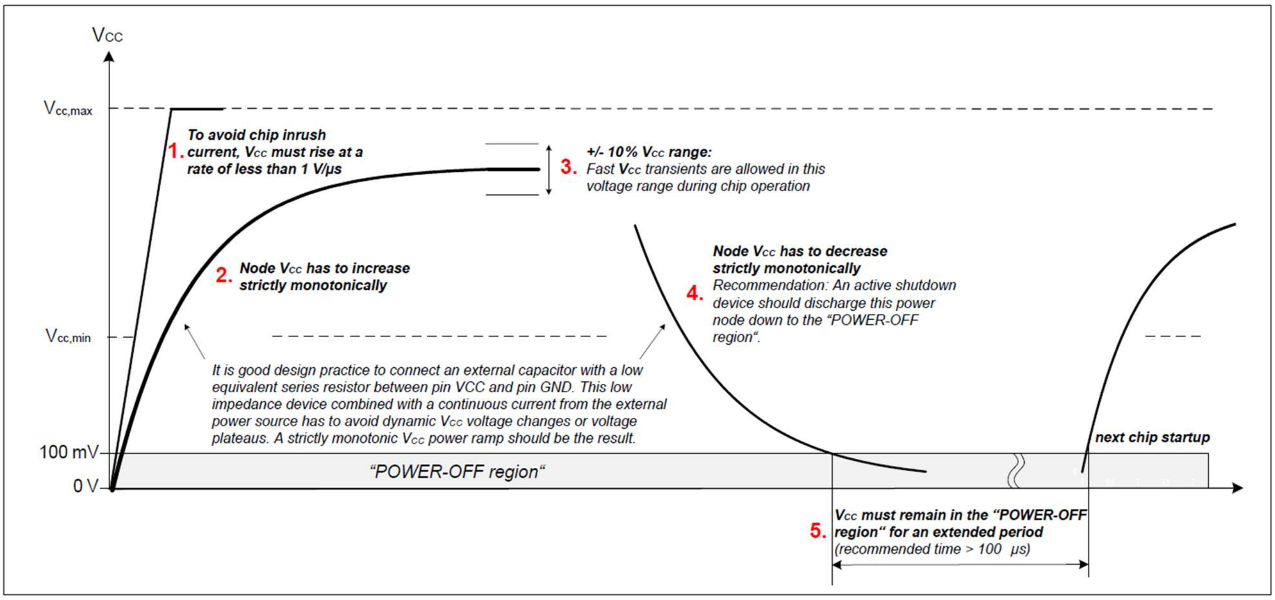
1) Bare die + typical package, e.g. VQFN or chipcard module. Package details are available on request.

- GPIO AC Electrical Characteristics

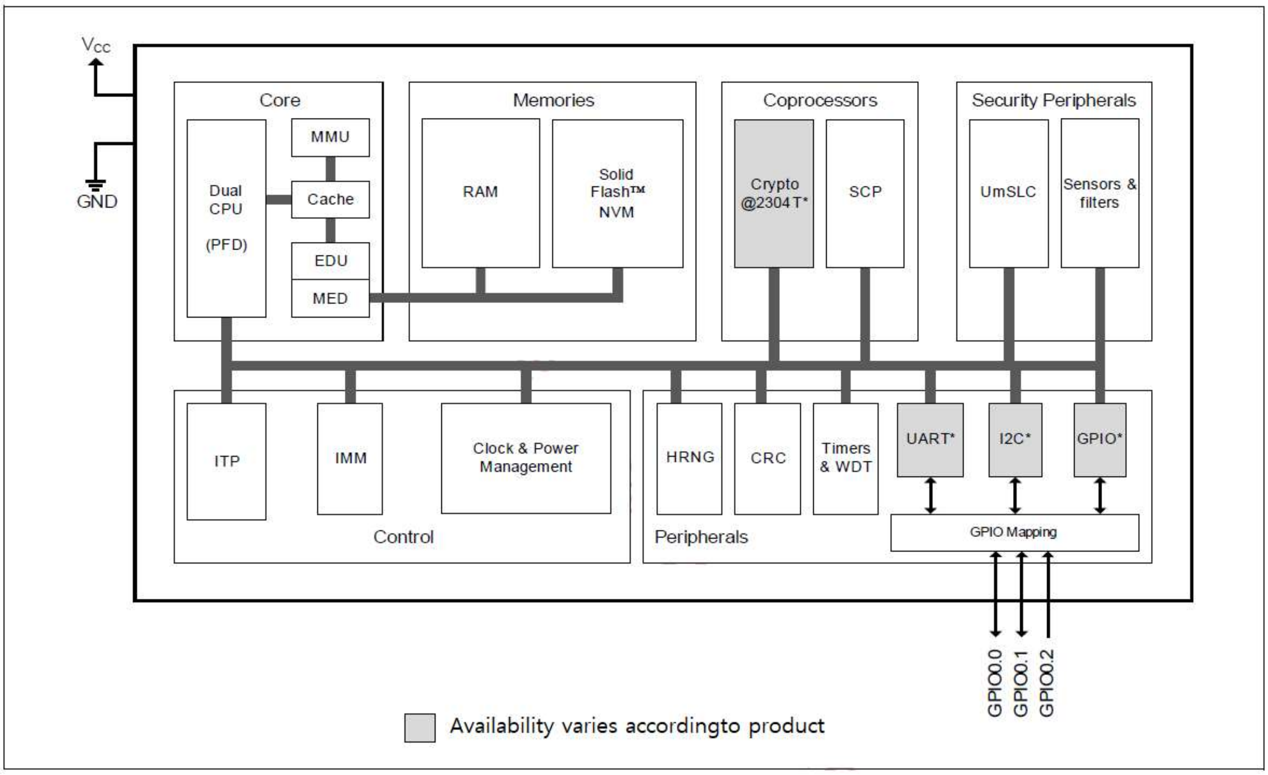
Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
Output signal rise time	t_r	–	–	10.0	ns	10% V_{CC_GPIO} to 90% V_{CC_GPIO}; C_{LOAD} = 15 pF, pull-up/down off, no DC load.
Output signal fall time	t_f	–	–	10.0	ns	90% V_{CC_GPIO} to 10% V_{CC_GPIO}; C_{LOAD} = 15 pF, pull-up/down off, no DC load.
GPIO input path low-pass filter	f_{CUTOFF}	20	–	40	MHz	50/50 duty cycle.
	t_{CUTOFF}[1)]	12.5	–	25	ns	High or low pulse width.

1) Spikes shorter than min. are filtered, spikes longer than max. are not filtered.

- Recommended Power-up Behavior



- Block Diagram



3.2 I2C Timing Diagram

3.2.1 I2C Active target characteristic

- I2C Standard Mode Interface Characteristics

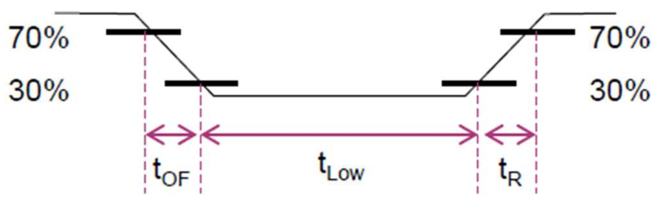
Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
SCL clock frequency	f_{SCL}	0	–	100	kHz	
Output fall time From V_{IHmin} to V_{ILmax} (at device pin)	T_{OF}	–	–	250	ns	$C_b \leq 400$ pF; $V_{cc\_I2C} \geq 2.7$ V $C_b \leq 200$ pF; $V_{cc\_I2C} < 2.7$ V

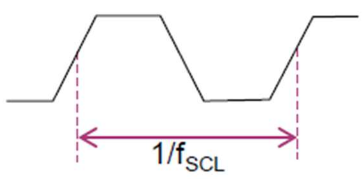
- I2C Fast Mode Interface Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
SCL clock frequency	f_{SCL}	0	–	400	kHz	
Output fall time From V_{IHmin} to V_{ILmax} (at device pin)	T_{OF}	20* V_{cc_I2C} / 5.5V	–	250	ns	$C_b \leq 400$ pF; $V_{cc\_I2C} \geq 2.7$ V $C_b \leq 200$ pF; $V_{cc\_I2C} < 2.7$ V

- I2C Fast Mode Plus Interface Characteristics

Parameter	Symbol	Values			Unit	Note or Test Condition
		Min.	Typ.	Max.		
SCL clock frequency	f_{SCL}	0	–	1000	kHz	
Output fall time From V_{IHmin} to V_{ILmax} (at device pin)	T_{OF}	20* V_{cc_I2C} / 5.5V	–	120	ns	$C_b \leq 150$ pF





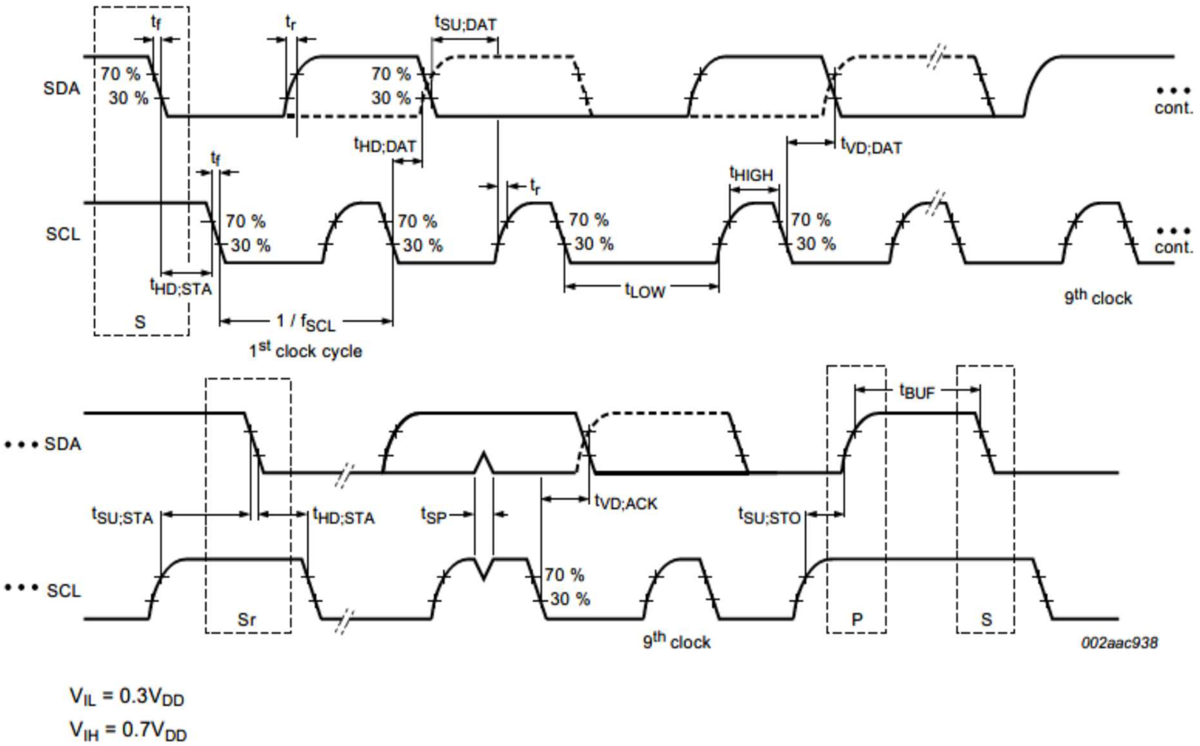
SCL & SCL

-. Rise/fall time 30%-70% (I2C)

-. t_R depends on external circuit(pull-up)

-. t_{Low} depends on f_{SCL}

3.2.2 I2C wave form diagram



Definition of timing for F/S-mode devices on the I²C-bus

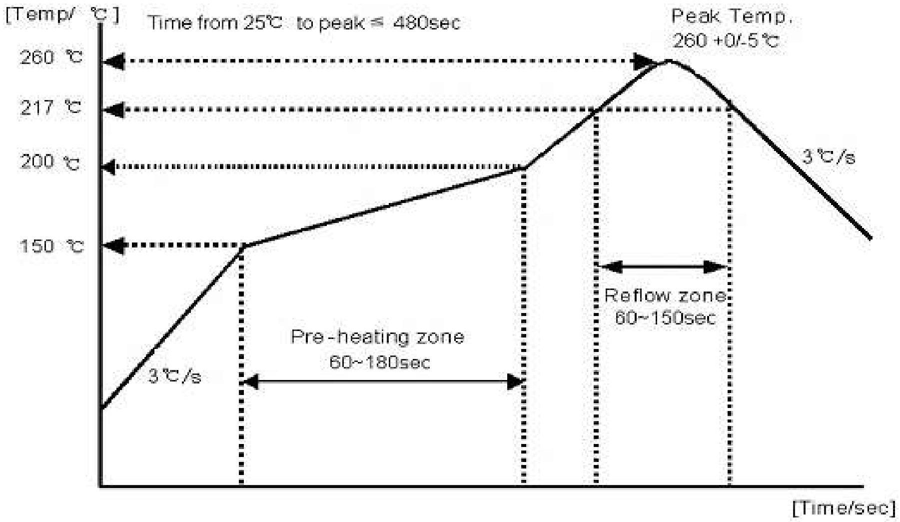
3.2.3 I2C startup and booting timing after VCC

▶ eSE startup and Booting Time(Typ.) : **6.857ms**

▶ Recommended delay timing from VCC(Including safety margin) : **20ms**

3.3 Reflow Profile

▶ Standard Reflow Condition (IPC / JEDEC Standard)



** Remark: IPC/JEDEC J-STD-020 Revision C (proposed standard for ballot January 2004)

Table 4-2 Pb-free Process - Package Peak Reflow Temperatures

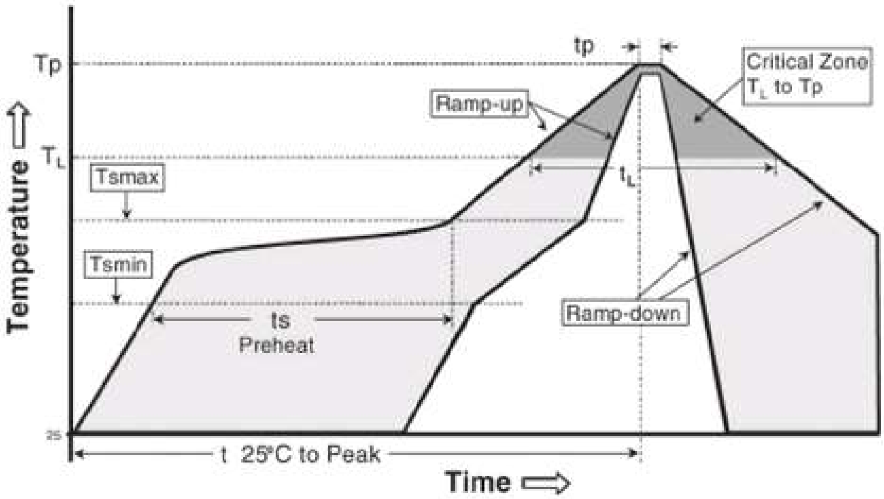
Package Thickness	Volume mm³ < 350	Volume mm³ 350 - 2000	Volume mm³ > 2000
< 1.6 mm	260 °C *	260 °C *	260 °C *
1.6 mm - 2.5 mm	260 °C *	250 °C *	245 °C *
> 2.5 mm	250 °C *	245 °C *	245 °C *
* Tolerance: The device manufacturer/supplier shall assure process compatibility up to and including the stated classification temperature at the rated MSL level			

Note 1: Package volume excludes external terminals (balls, bumps, lands, leads) and/or non-integral heat sinks.

Note 2: The maximum component temperature reached during reflow depends on package thickness and volume. The use of convection reflow processes reduces the thermal gradients between packages. However, thermal gradients due to differences in thermal mass of SMD packages may still exist.

Profile Feature	Sn-Pb Eutectic Assembly	Pb-Free Assembly
Average ramp-up rate (Ts_{max} to Tp)	3° C/second max.	3° C/second max.
Preheat		
- Temperature Min (Ts_{min})	100 °C	150 °C
- Temperature Max (Ts_{max})	150 °C	200 °C
- Time (Ts_{min} to Ts_{max}) (ts)	60-120 seconds	60-180 seconds
Time maintained above:		
- Temperature (T_L)	183 °C	217 °C
- Time (t_L)	60-150 seconds	60-150 seconds
Peak Temperature (Tp)	See Table 4.1	See Table 4.2
Time within 5°C of actual Peak Temperature (tp)[2]	10-30 seconds	20-40 seconds
Ramp-down Rate	6 °C/second max.	6 °C/second max.
Time 25°C to Peak Temperature	6 minutes max.	8 minutes max.

Note 1: All temperatures refer to topside of the package, measured on the package body surface.
Note 2: Time within 5 °C of actual peak temperature (tp) specified for the reflow profiles is a "supplier" minimum and "user" maximum.



▶ Reflow Specification(CTQ) and warranty

CTQ Process Control Point		Spec	
		LSL	USL
Reflow	Temperature Range[℃]	240	250
	Continuous time 220℃ more [sec]	30	60

▶ 수동 납땜 (납땜 인두기를 사용할 경우)

① 예 열: 120℃
② 시 간: 60 ~ 300 sec.
③ 인두온도: 340℃±5℃
④ 시 간: 각 단 최대 5 sec.